LAPORAN PRAKTIKUM

POSTTEST (5)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

Khayzan Dwitra Attala (2509106103)

KELAS (C'1)

PROGRAM STUDI INFORMATIKA

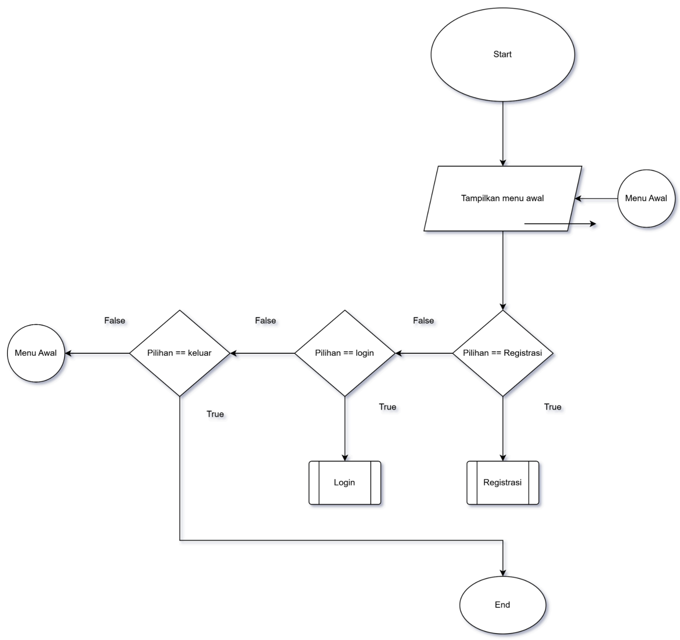
UNIVERSITAS MULAWARMAN

SAMARINDA

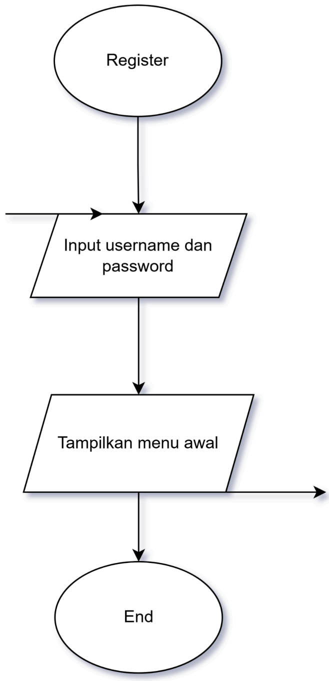
2025

1. Flowchart

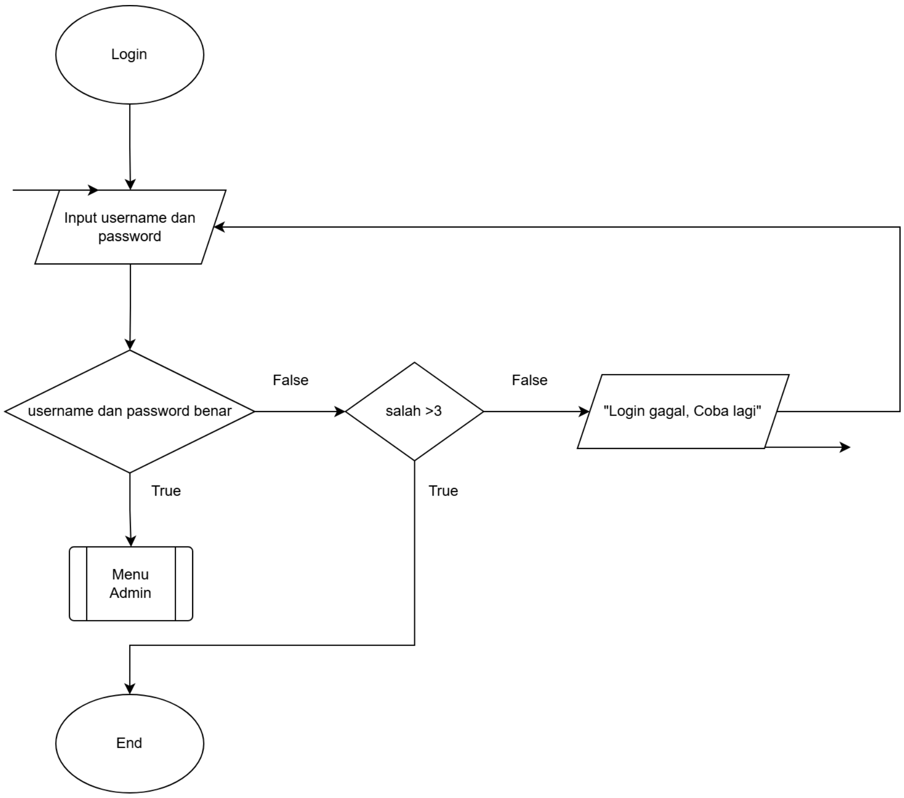
Flowchart ini membantu memvisualisasikan keputusan bercabang (pakai simbol belah ketupat) dan tindakan proses (pakai persegi panjang), sehingga mudah melihat jalur mana yang ditempuh sesuai kondisi.. Kemudian flowchart menggunakan predefined proses agar menunjukkan bahwa ada proses/fungsi yang terdefinisikan



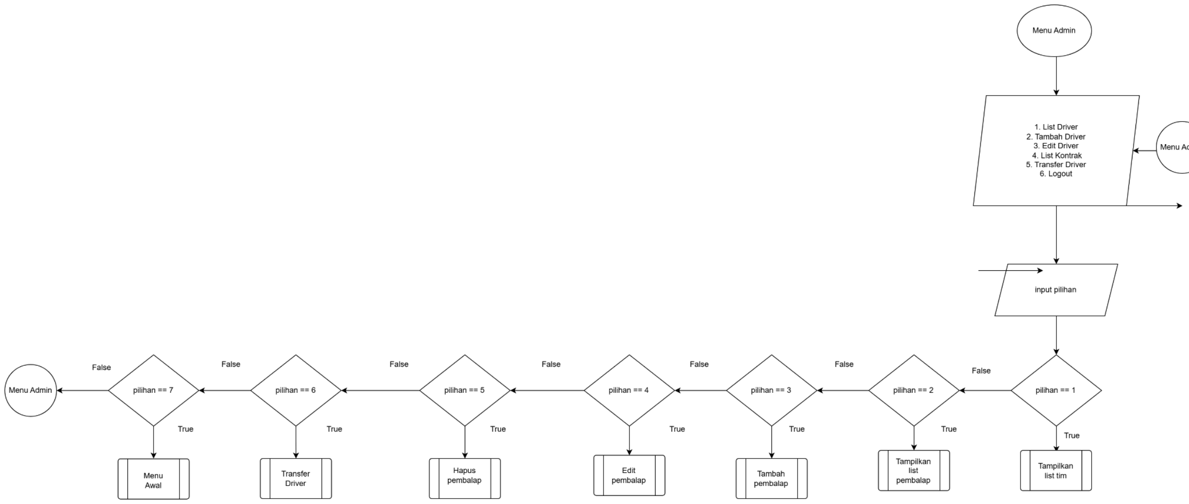
Gambar 1.1 Flowchart CRUD Menu awal



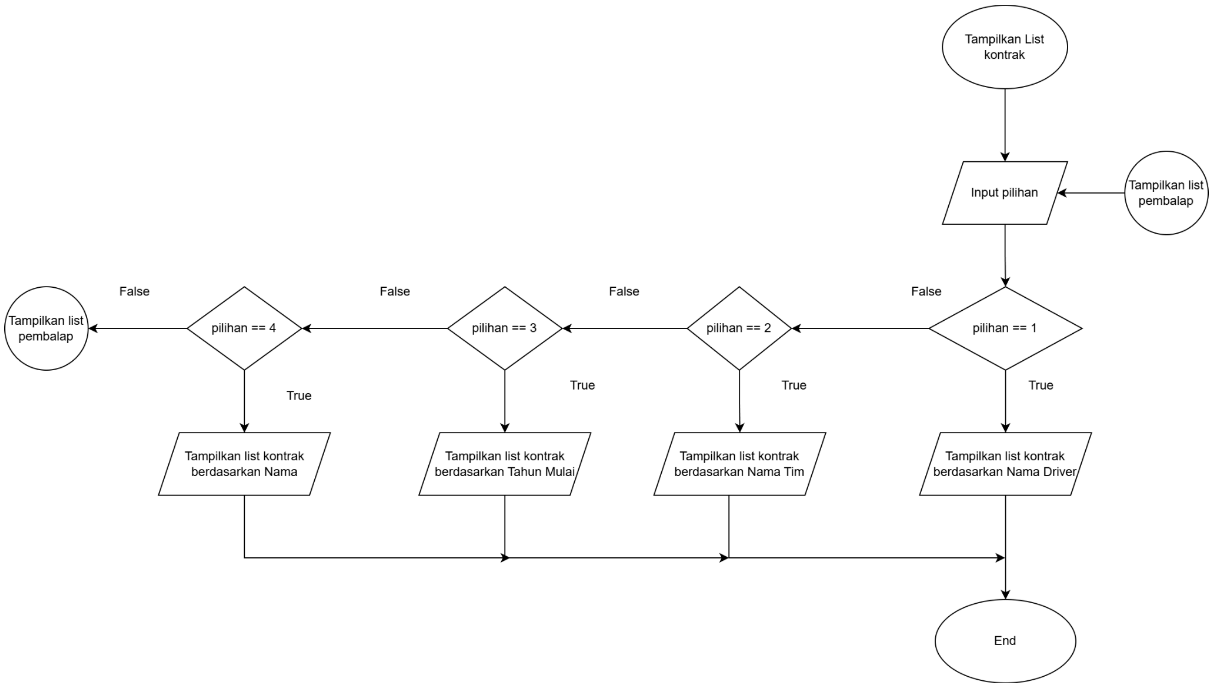
Gambar 1.2 Flowchart Register



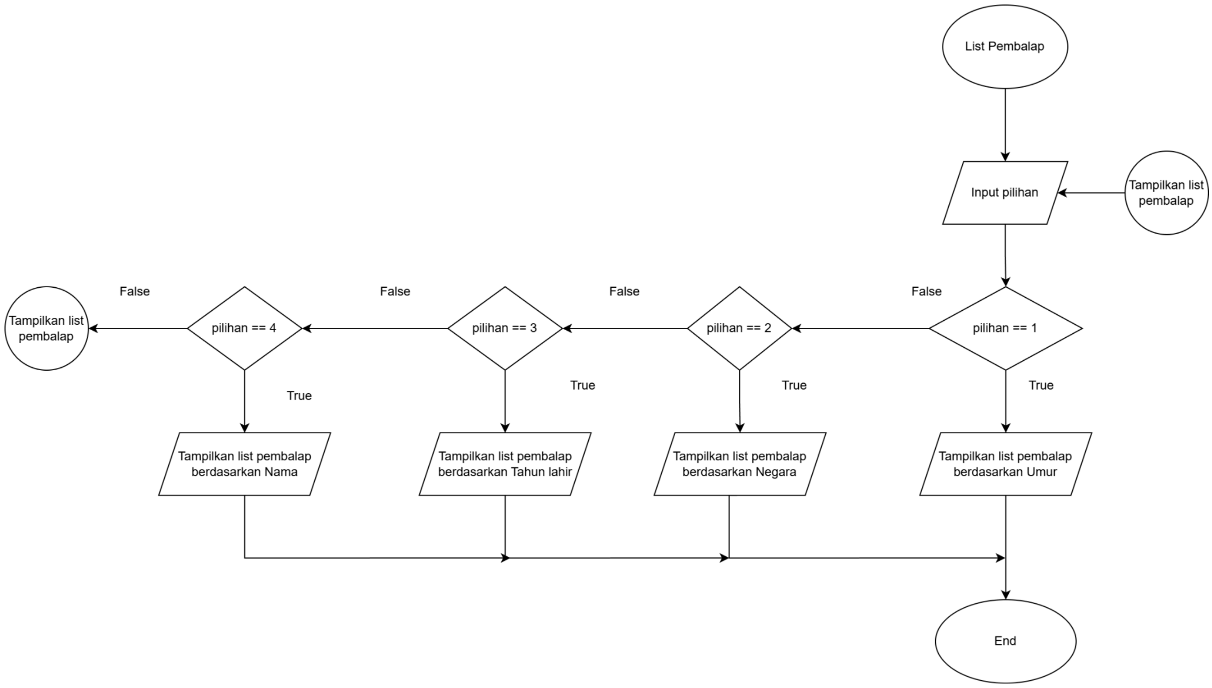
Gambar 1.3 Flowchart Login



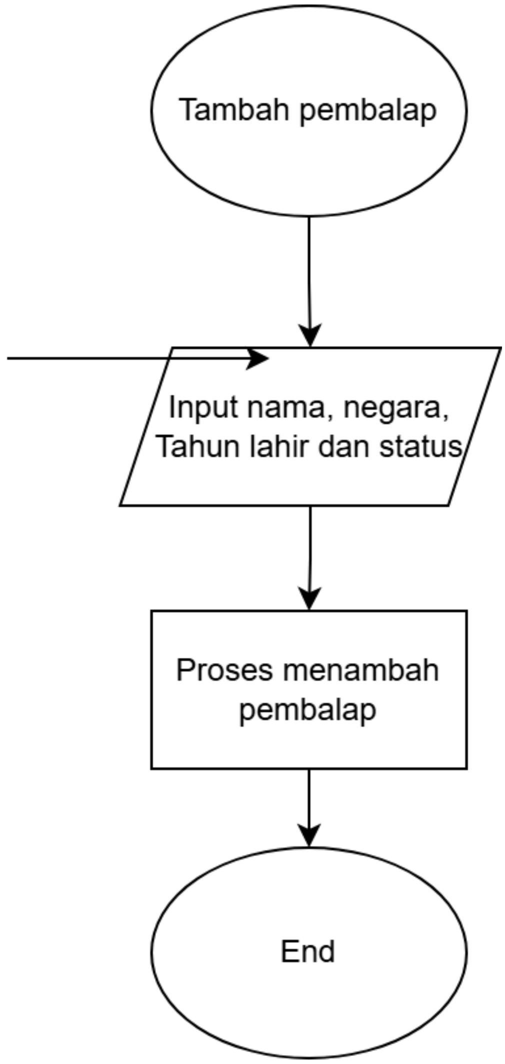
Gambar 1.4 Flowchart Menu Admin



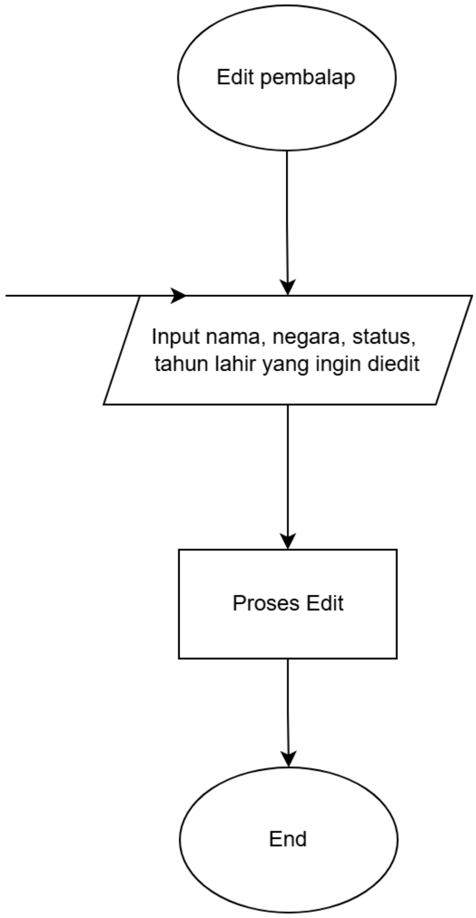
Gambar 1.5 Tampilkan List Kontrak



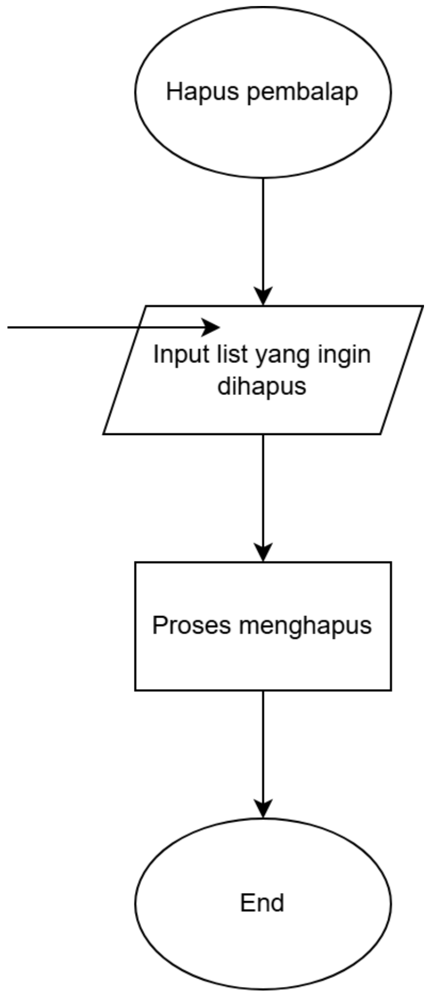
Gambar 1.6 Tampilkan list pembalap



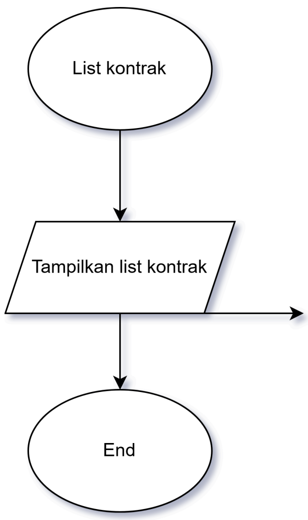
Gambar 1.7 Tambah pembalap



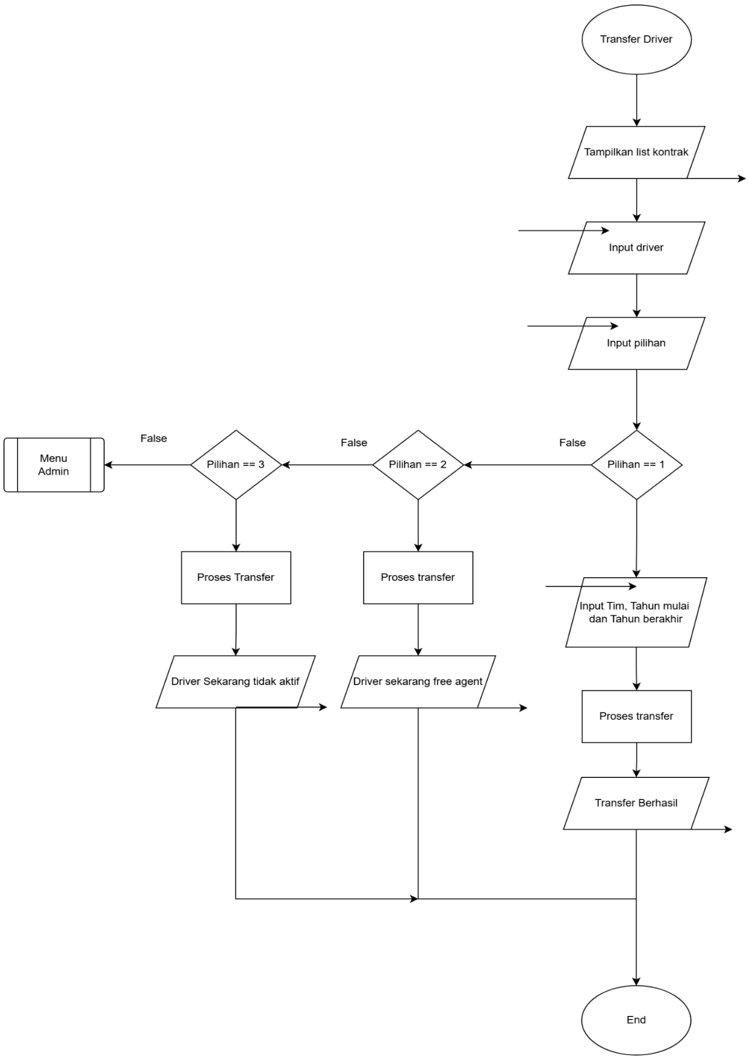
Gambar 1.8 Edit Pembalap



Gambar 1.9 Hapus pembalap



Gambar 1.10 List Kontrak



Gambar 1.11 Transfer Driver

2. Deskripsi Singkat Program

Secara keseluruhan, program ini berfungsi sebagai sistem manajemen kontrak dan informasi pembalap F1 yang sederhana, dengan kontrol akses berdasarkan role dan menu interaktif di konsol.

3. Source Code

```cpp
#include <iostream>
#include <string>
#include <iomanip>
#include <utility>
#include <limits>

using namespace std;

const int MAX_DRIVERS = 50;
const int MAX_USERS = 20;
const int MAX_TEAMS = 11;
const int MAX_CONTRACTS = 100;
const int CURRENT_YEAR = 2026;

struct Driver {
    string name;
    string nationality;
    int birthYear;
    string status;
};

struct User {
    string username;
    string password;
};

struct Team {
    string name;
};

struct Contract {
    int driverIndex;
    int teamIndex;
    int startYear;
    int endYear;
};

void clearInput() {
    cin.clear();
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
}
```

```cpp
bool isValidDriverIndex(int index, int totalDrivers) {
    return index >= 0 && index < totalDrivers;
}

bool isValidTeamIndex(int index, int totalTeams) {
    return index >= 0 && index < totalTeams;
}

bool isContractActiveInYear(const Contract &contract, int year) {
    return contract.startYear <= year && contract.endYear >= year;
}


int partitionName(Driver drivers[], int low, int high) {
    string pivot = drivers[high].name;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (drivers[j].name < pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByName(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionName(drivers, low, high);
        quickSortByName(drivers, low, pivot - 1);
        quickSortByName(drivers, pivot + 1, high);
    }
}

int partitionNationality(Driver drivers[], int low, int high) {
    string pivot = drivers[high].nationality;
    int i = low - 1;
```

```
    for (int j = low; j < high; j++) {
        if (drivers[j].nationality < pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByNationality(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionNationality(drivers, low, high);
        quickSortByNationality(drivers, low, pivot - 1);
        quickSortByNationality(drivers, pivot + 1, high);
    }
}

int partitionBirthYear(Driver drivers[], int low, int high) {
    int pivot = drivers[high].birthYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (drivers[j].birthYear > pivot) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByBirthYear(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionBirthYear(drivers, low, high);
        quickSortByBirthYear(drivers, low, pivot - 1);
        quickSortByBirthYear(drivers, pivot + 1, high);
    }
}
```

```
int partitionAge(Driver drivers[], int low, int high) {
    int pivotAge = CURRENT_YEAR - drivers[high].birthYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        int ageJ = CURRENT_YEAR - drivers[j].birthYear;
        if (ageJ > pivotAge) {
            i++;
            swap(drivers[i], drivers[j]);
        }
    }

    swap(drivers[i + 1], drivers[high]);
    return i + 1;
}

void quickSortByAge(Driver drivers[], int low, int high) {
    if (low < high) {
        int pivot = partitionAge(drivers, low, high);
        quickSortByAge(drivers, low, pivot - 1);
        quickSortByAge(drivers, pivot + 1, high);
    }
}


int partitionContractByDriverName(Contract contracts[], int low, int high,
Driver drivers[]) {
    string pivot = drivers[contracts[high].driverIndex].name;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (drivers[contracts[j].driverIndex].name < pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
}
```

```
void quickSortContractByDriverName(Contract contracts[], int low, int high,
Driver drivers[]) {
    if (low < high) {
        int pivot = partitionContractByDriverName(contracts, low, high,
drivers);
        quickSortContractByDriverName(contracts, low, pivot - 1, drivers);
        quickSortContractByDriverName(contracts, pivot + 1, high, drivers);
    }
}

int partitionContractByTeamName(Contract contracts[], int low, int high,
Team teams[]) {
    string pivot = teams[contracts[high].teamIndex].name;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (teams[contracts[j].teamIndex].name < pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
}

void quickSortContractByTeamName(Contract contracts[], int low, int high,
Team teams[]) {
    if (low < high) {
        int pivot = partitionContractByTeamName(contracts, low, high,
teams);
        quickSortContractByTeamName(contracts, low, pivot - 1, teams);
        quickSortContractByTeamName(contracts, pivot + 1, high, teams);
    }
}

int partitionContractByStartYear(Contract contracts[], int low, int high) {
    int pivot = contracts[high].startYear;
    int i = low - 1;
```

```
    for (int j = low; j < high; j++) {
        if (contracts[j].startYear > pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
}

void quickSortContractByStartYear(Contract contracts[], int low, int high) {
    if (low < high) {
        int pivot = partitionContractByStartYear(contracts, low, high);
        quickSortContractByStartYear(contracts, low, pivot - 1);
        quickSortContractByStartYear(contracts, pivot + 1, high);
    }
}

int partitionContractByEndYear(Contract contracts[], int low, int high) {
    int pivot = contracts[high].endYear;
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (contracts[j].endYear > pivot) {
            i++;
            swap(contracts[i], contracts[j]);
        }
    }

    swap(contracts[i + 1], contracts[high]);
    return i + 1;
}

void quickSortContractByEndYear(Contract contracts[], int low, int high) {
    if (low < high) {
        int pivot = partitionContractByEndYear(contracts, low, high);
        quickSortContractByEndYear(contracts, low, pivot - 1);
        quickSortContractByEndYear(contracts, pivot + 1, high);
    }
}
```

```
void initializeDrivers(Driver drivers[], int &totalDrivers) {
    string names[] = {
        "Charles Leclerc", "Lewis Hamilton", "George Russell", "Kimi
Antonelli",
        "Lando Norris", "Oscar Piastri", "Fernando Alonso", "Lance Stroll",
        "Esteban Ocon", "Oliver Bearman", "Max Verstappen", "Sergio Perez",
        "Arvid Lindblad", "Pierre Gasly", "Valtteri Bottas", "Isack Hadjar",
        "Nico Hulkenberg", "Gabriel Bortoletto", "Liam Lawson", "Alexander
Albon",
        "Carlos Sainz", "Franco Colapinto"
    };

    string countries[] = {
        "Monaco", "UK", "UK", "Italy",
        "UK", "Australia", "Spain", "Canada",
        "France", "UK", "Netherlands", "Mexico",
        "Sweden", "France", "Finland", "France",
        "Germany", "Brazil", "New Zealand", "Thailand",
        "Spain", "Argentina"
    };

    int birthYears[] = {
        1997, 1985, 1998, 2006,
        1999, 2001, 1981, 1998,
        1996, 2005, 1997, 1990,
        2004, 1996, 1989, 2003,
        1987, 2004, 2002, 1996,
        1994, 2003
    };

    for (int i = 0; i < 22; i++) {
        drivers[i].name = names[i];
        drivers[i].nationality = countries[i];
        drivers[i].birthYear = birthYears[i];
        drivers[i].status = "Aktif";
    }

    totalDrivers = 22;
}
```

```cpp
void initializeTeams(Team teams[], int &totalTeams) {
    string names[] = {
        "Ferrari", "Mercedes", "McLaren", "Red Bull", "RB",
        "Alpine", "Aston Martin", "Audi", "Williams", "Cadillac", "Haas"
    };

    for (int i = 0; i < 11; i++) {
        teams[i].name = names[i];
    }

    totalTeams = 11;
}

void initializeUsers(User users[], int &totalUsers) {
    users[0].username = "Attol";
    users[0].password = "2509106103";
    totalUsers = 1;
}

void initializeContracts(Contract contracts[], int &totalContracts) {
    int map[22] = {0, 0, 1, 1, 2, 2, 6, 6, 10, 10, 3, 9, 4, 5, 9, 3, 7, 7,
4, 8, 8, 5};

    for (int i = 0; i < 22; i++) {
        contracts[i].driverIndex = i;
        contracts[i].teamIndex = map[i];
        contracts[i].startYear = 2025;
        contracts[i].endYear = 2027;
    }

    totalContracts = 22;
}


void registerUser(User users[], int &totalUsers) {
    if (totalUsers >= MAX_USERS) {
        cout << "Kapasitas user penuh!\n";
        return;
    }

    cout << "Username: ";
```

```
    getline(cin, users[totalUsers].username);

    cout << "Password: ";
    getline(cin, users[totalUsers].password);

    totalUsers++;
    cout << "Registrasi berhasil!\n";
}

bool loginUser(User users[], int totalUsers) {
    string username, password;

    cout << "Username: ";
    getline(cin, username);

    cout << "Password: ";
    getline(cin, password);

    for (int i = 0; i < totalUsers; i++) {
        if (users[i].username == username && users[i].password == password)
{
            return true;
        }
    }

    cout << "Login gagal!\n";
    return false;
}


void displayDrivers(Driver drivers[], int totalDrivers) {
    cout << "\n=== LIST DRIVER ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Nama"
         << setw(15) << "Negara"
         << setw(12) << "Tahun Lahir"
         << setw(8)  << "Umur"
         << setw(15) << "Status" << '\n';

    cout << string(79, '=') << '\n';
```

```
    for (int i = 0; i < totalDrivers; i++) {
        cout << left
             << setw(4)  << i + 1
             << setw(25) << drivers[i].name
             << setw(15) << drivers[i].nationality
             << setw(12) << drivers[i].birthYear
             << setw(8)  << (CURRENT_YEAR - drivers[i].birthYear)
             << setw(15) << drivers[i].status
             << '\n';
    }
}

void displayTeams(Team teams[], int totalTeams) {
    cout << "\n=== LIST TEAM ===\n";
    for (int i = 0; i < totalTeams; i++) {
        cout << i + 1 << ". " << teams[i].name << '\n';
    }
}

void displayAllContracts(Contract contracts[], int totalContracts, Driver
drivers[], Team teams[]) {
    cout << "\n=== LIST SEMUA KONTRAK ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Driver"
         << setw(20) << "Tim"
         << setw(12) << "Mulai"
         << setw(12) << "Selesai"
         << setw(15) << "Status 2026"
         << '\n';

    cout << string(88, '=') << '\n';

    for (int i = 0; i < totalContracts; i++) {
        string status2026 = isContractActiveInYear(contracts[i],
CURRENT_YEAR) ? "Aktif" : "Tidak Aktif";

        cout << left
             << setw(4)  << i + 1
             << setw(25) << drivers[contracts[i].driverIndex].name
```

```
            << setw(20) << teams[contracts[i].teamIndex].name
            << setw(12) << contracts[i].startYear
            << setw(12) << contracts[i].endYear
            << setw(15) << status2026
            << '\n';
    }

    if (totalContracts == 0) {
        cout << "Belum ada data kontrak.\n";
    }
}

void displayActiveContracts2026(Contract contracts[], int totalContracts,
Driver drivers[], Team teams[]) {
    cout << "\n=== LIST KONTRAK AKTIF TAHUN " << CURRENT_YEAR << " ===\n";
    cout << left
         << setw(4)  << "No"
         << setw(25) << "Driver"
         << setw(20) << "Tim"
         << setw(12) << "Mulai"
         << setw(12) << "Selesai"
         << '\n';

    cout << string(73, '=') << '\n';

    int nomor = 1;
    for (int i = 0; i < totalContracts; i++) {
        if (isContractActiveInYear(contracts[i], CURRENT_YEAR)) {
            cout << left
                 << setw(4)  << nomor
                 << setw(25) << drivers[contracts[i].driverIndex].name
                 << setw(20) << teams[contracts[i].teamIndex].name
                 << setw(12) << contracts[i].startYear
                 << setw(12) << contracts[i].endYear
                 << '\n';
            nomor++;
        }
    }

    if (nomor == 1) {
        cout << "Tidak ada kontrak aktif pada tahun " << CURRENT_YEAR <<
```

```cpp
".\n";
    }
}


void sortDriversMenu(Driver drivers[], int totalDrivers) {
    int sortChoice;

    cout << "\n=== SORT DRIVER BY ===\n";
    cout << "1. Umur (Descending)\n";
    cout << "2. Negara (A-Z)\n";
    cout << "3. Tahun lahir (Descending)\n";
    cout << "4. Nama (A-Z)\n";
    cout << "Pilih: ";
    cin >> sortChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (sortChoice) {
        case 1:
            quickSortByAge(drivers, 0, totalDrivers - 1);
            break;
        case 2:
            quickSortByNationality(drivers, 0, totalDrivers - 1);
            break;
        case 3:
            quickSortByBirthYear(drivers, 0, totalDrivers - 1);
            break;
        case 4:
            quickSortByName(drivers, 0, totalDrivers - 1);
            break;
        default:
            cout << "Pilihan sort tidak valid!\n";
            return;
    }

    displayDrivers(drivers, totalDrivers);
```

```
}


void sortContractsMenu(Contract contracts[], int totalContracts, Driver
drivers[], Team teams[]) {
    int sortChoice;
    int filterChoice;

    cout << "\n=== SORT KONTRAK BY ===\n";
    cout << "1. Nama Driver (A-Z)\n";
    cout << "2. Nama Tim (A-Z)\n";
    cout << "3. Tahun Mulai (Descending)\n";
    cout << "4. Tahun Selesai (Descending)\n";
    cout << "Pilih: ";
    cin >> sortChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (sortChoice) {
        case 1:
            quickSortContractByDriverName(contracts, 0, totalContracts - 1,
drivers);
            break;
        case 2:
            quickSortContractByTeamName(contracts, 0, totalContracts - 1,
teams);
            break;
        case 3:
            quickSortContractByStartYear(contracts, 0, totalContracts - 1);
            break;
        case 4:
            quickSortContractByEndYear(contracts, 0, totalContracts - 1);
            break;
        default:
            cout << "Pilihan sort tidak valid!\n";
            return;
    }
```

```
    cout << "\n=== TAMPILKAN KONTRAK ===\n";
    cout << "1. Semua kontrak\n";
    cout << "2. Hanya kontrak aktif tahun 2026\n";
    cout << "Pilih: ";
    cin >> filterChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    switch (filterChoice) {
        case 1:
            displayAllContracts(contracts, totalContracts, drivers, teams);
            break;
        case 2:
            displayActiveContracts2026(contracts, totalContracts, drivers,
teams);
            break;
        default:
            cout << "Pilihan tampilan tidak valid!\n";
            break;
    }
}


void addDriver(Driver drivers[], int &totalDrivers) {
    if (totalDrivers >= MAX_DRIVERS) {
        cout << "Data driver penuh!\n";
        return;
    }

    clearInput();

    cout << "Nama: ";
    getline(cin, drivers[totalDrivers].name);

    cout << "Negara: ";
    getline(cin, drivers[totalDrivers].nationality);
```

```cpp
    cout << "Tahun lahir: ";
    cin >> drivers[totalDrivers].birthYear;

    if (cin.fail()) {
        clearInput();
        cout << "Input tahun lahir tidak valid!\n";
        return;
    }

    clearInput();

    cout << "Status: ";
    getline(cin, drivers[totalDrivers].status);

    totalDrivers++;
    cout << "Driver berhasil ditambahkan!\n";
}


void editDriver(Driver drivers[], int totalDrivers) {
    if (totalDrivers == 0) {
        cout << "Tidak ada data driver.\n";
        return;
    }

    displayDrivers(drivers, totalDrivers);

    int choice;
    cout << "Pilih nomor driver yang ingin diedit: ";
    cin >> choice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    int index = choice - 1;
    if (!isValidDriverIndex(index, totalDrivers)) {
        cout << "Nomor driver tidak valid!\n";
```

```
        return;
    }

    clearInput();

    cout << "Nama baru: ";
    getline(cin, drivers[index].name);

    cout << "Negara baru: ";
    getline(cin, drivers[index].nationality);

    cout << "Tahun lahir baru: ";
    cin >> drivers[index].birthYear;

    if (cin.fail()) {
        clearInput();
        cout << "Input tahun lahir tidak valid!\n";
        return;
    }

    clearInput();

    cout << "Status baru: ";
    getline(cin, drivers[index].status);

    cout << "Data driver berhasil diupdate!\n";
}


void deleteDriverContracts(Contract contracts[], int &totalContracts, int
driverIndex) {
    for (int i = 0; i < totalContracts; i++) {
        if (contracts[i].driverIndex == driverIndex) {
            for (int j = i; j < totalContracts - 1; j++) {
                contracts[j] = contracts[j + 1];
            }
            totalContracts--;
            i--;
        }
    }
}
```

```
void adjustContractDriverIndexes(Contract contracts[], int totalContracts,
int deletedDriverIndex) {
    for (int i = 0; i < totalContracts; i++) {
        if (contracts[i].driverIndex > deletedDriverIndex) {
            contracts[i].driverIndex--;
        }
    }
}

void deleteDriver(Driver drivers[], int &totalDrivers, Contract contracts[],
int &totalContracts) {
    if (totalDrivers == 0) {
        cout << "Tidak ada data driver.\n";
        return;
    }

    displayDrivers(drivers, totalDrivers);

    int choice;
    cout << "Pilih nomor driver yang ingin dihapus: ";
    cin >> choice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tidak valid!\n";
        return;
    }

    int index = choice - 1;
    if (!isValidDriverIndex(index, totalDrivers)) {
        cout << "Nomor driver tidak valid!\n";
        return;
    }

    deleteDriverContracts(contracts, totalContracts, index);
    adjustContractDriverIndexes(contracts, totalContracts, index);

    for (int i = index; i < totalDrivers - 1; i++) {
        drivers[i] = drivers[i + 1];
    }
```

```
    totalDrivers--;
    cout << "Driver berhasil dihapus!\n";
}


void addContract(Driver drivers[], int totalDrivers,
                 Team teams[], int totalTeams,
                 Contract contracts[], int &totalContracts) {
    if (totalContracts >= MAX_CONTRACTS) {
        cout << "Data kontrak penuh!\n";
        return;
    }

    int driverChoice, teamChoice, startYear, endYear;

    displayDrivers(drivers, totalDrivers);
    cout << "Pilih driver: ";
    cin >> driverChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input driver tidak valid!\n";
        return;
    }

    int driverIndex = driverChoice - 1;
    if (!isValidDriverIndex(driverIndex, totalDrivers)) {
        cout << "Pilihan driver tidak valid!\n";
        return;
    }

    displayTeams(teams, totalTeams);
    cout << "Pilih tim: ";
    cin >> teamChoice;

    if (cin.fail()) {
        clearInput();
        cout << "Input tim tidak valid!\n";
        return;
    }
```

```
    int teamIndex = teamChoice - 1;
    if (!isValidTeamIndex(teamIndex, totalTeams)) {
        cout << "Pilihan tim tidak valid!\n";
        return;
    }

    cout << "Tahun mulai: ";
    cin >> startYear;
    if (cin.fail()) {
        clearInput();
        cout << "Input tahun mulai tidak valid!\n";
        return;
    }

    cout << "Tahun selesai: ";
    cin >> endYear;
    if (cin.fail()) {
        clearInput();
        cout << "Input tahun selesai tidak valid!\n";
        return;
    }

    if (endYear < startYear) {
        cout << "Tahun selesai tidak boleh lebih kecil dari tahun mulai!\n";
        return;
    }

    contracts[totalContracts].driverIndex = driverIndex;
    contracts[totalContracts].teamIndex = teamIndex;
    contracts[totalContracts].startYear = startYear;
    contracts[totalContracts].endYear = endYear;
    totalContracts++;

    drivers[driverIndex].status = "Aktif";

    cout << "Kontrak berhasil ditambahkan!\n";
}


void transferDriver(Driver drivers[], int totalDrivers,
```

```cpp
                    Team teams[], int totalTeams,
                    Contract contracts[], int &totalContracts) {
    addContract(drivers, totalDrivers, teams, totalTeams, contracts,
totalContracts);
}


void adminMenu(Driver drivers[], int &totalDrivers,
               Team teams[], int totalTeams,
               Contract contracts[], int &totalContracts) {
    int choice;

    do {
        cout << "\n=== MENU ADMIN ===\n";
        cout << "1. List Driver\n";
        cout << "2. Tambah Driver\n";
        cout << "3. Edit Driver\n";
        cout << "4. Hapus Driver\n";
        cout << "5. List Kontrak\n";
        cout << "6. Tambah / Transfer Kontrak\n";
        cout << "7. Logout\n";
        cout << "Pilih: ";
        cin >> choice;

        if (cin.fail()) {
            clearInput();
            cout << "Input tidak valid!\n";
            continue;
        }

        switch (choice) {
            case 1:
                sortDriversMenu(drivers, totalDrivers);
                break;
            case 2:
                addDriver(drivers, totalDrivers);
                break;
            case 3:
                editDriver(drivers, totalDrivers);
                break;
            case 4:
```

```
                deleteDriver(drivers, totalDrivers, contracts,
totalContracts);
                break;
            case 5:
                if (totalContracts > 0) {
                    sortContractsMenu(contracts, totalContracts, drivers,
teams);
                } else {
                    cout << "Tidak ada data kontrak.\n";
                }
                break;
            case 6:
                transferDriver(drivers, totalDrivers, teams, totalTeams,
contracts, totalContracts);
                break;
            case 7:
                cout << "Logout...\n";
                break;
            default:
                cout << "Menu belum tersedia.\n";
        }

    } while (choice != 7);
}


int main() {
    Driver drivers[MAX_DRIVERS];
    User users[MAX_USERS];
    Team teams[MAX_TEAMS];
    Contract contracts[MAX_CONTRACTS];

    int totalDrivers = 0;
    int totalUsers = 0;
    int totalTeams = 0;
    int totalContracts = 0;

    initializeDrivers(drivers, totalDrivers);
    initializeTeams(teams, totalTeams);
    initializeUsers(users, totalUsers);
    initializeContracts(contracts, totalContracts);
```

```
    int choice;

    do {
        cout << "\n=== MENU AWAL ===\n";
        cout << "1. Register\n";
        cout << "2. Login\n";
        cout << "3. Keluar\n";
        cout << "Pilih: ";
        cin >> choice;

        if (cin.fail()) {
            clearInput();
            cout << "Input tidak valid!\n";
            continue;
        }

        clearInput();

        if (choice == 1) {
            registerUser(users, totalUsers);
        } else if (choice == 2) {
            if (loginUser(users, totalUsers)) {
                adminMenu(drivers, totalDrivers, teams, totalTeams,
contracts, totalContracts);
            }
        } else if (choice == 3) {
            cout << "Program selesai.\n";
        } else {
            cout << "Pilihan tidak valid.\n";
        }

    } while (choice != 3);

    return 0;
}
```

4. Hasil Output

```
+-------------------------------------------+
|      SELAMAT DATANG DI BURSA TRANSFER F1  |
+-------------------------------------------+


==================================================
MENU AWAL
==================================================
+-----------------------------------+
| 1.  Register Admin                |
| 2.  Login Admin                   |
| 3.  Keluar                        |
+-----------------------------------+
Pilihan : 2
Username : Attol
Password : 2509106103
Login berhasil.
```

Gambar 4.1 Hasil Output Menu Awal

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih:
```

Gambar 4.2 Hasil Output Login

```
================================================================================
DATA PEMBALAP LENGKAP
================================================================================
+----+------------------------+---------------+------------+-------+------------+
| No | Nama                   | Negara        | Tahun Lahir | Umur  | Status     |
+----+------------------------+---------------+------------+-------+------------+
| 1  | Charles Leclerc        | Monaco        | 1997       | 29    | Aktif      |
| 2  | Lewis Hamilton         | UK            | 1985       | 41    | Aktif      |
| 3  | George Russell         | UK            | 1998       | 28    | Aktif      |
| 4  | Kimi Antonelli         | Italy         | 2006       | 20    | Aktif      |
| 5  | Lando Norris           | UK            | 1999       | 27    | Aktif      |
| 6  | Oscar Piastri          | Australia     | 2001       | 25    | Aktif      |
| 7  | Fernando Alonso        | Spain         | 1981       | 45    | Aktif      |
| 8  | Lance Stroll           | Canada        | 1998       | 28    | Aktif      |
| 9  | Esteban Ocon           | France        | 1996       | 30    | Aktif      |
| 10 | Oliver Bearman         | UK            | 2005       | 21    | Aktif      |
| 11 | Isack Hadjar           | France        | 2003       | 23    | Aktif      |
| 12 | Max Verstappen         | Netherlands   | 1997       | 29    | Aktif      |
| 13 | Carlos Sainz Jr.       | Spain         | 1994       | 32    | Aktif      |
| 14 | Alexander Albon        | UK            | 1996       | 30    | Aktif      |
| 15 | Valtteri Bottas        | Finland       | 1989       | 37    | Aktif      |
| 16 | Sergio Perez           | Mexico        | 1990       | 36    | Aktif      |
| 17 | Nico Hulkenberg        | Germany       | 1987       | 39    | Aktif      |
| 18 | Gabriel Bortoleto      | Brazil        | 2004       | 22    | Aktif      |
| 19 | Pierre Gasly           | France        | 1996       | 30    | Aktif      |
| 20 | Franco Colapinto       | Argentina     | 1999       | 27    | Aktif      |
| 21 | Liam Lawson            | New Zealand   | 2002       | 24    | Aktif      |
| 22 | Arvid Lindblad         | Sweden        | 2004       | 22    | Aktif      |
+----+------------------------+---------------+------------+-------+------------+
```

Gambar 4.3 Hasil Output List pembalap

```
=== MENU ===
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Transfer Driver
7. Logout
Pilih: 2
Nama: Benjamin Seskowi
Negara: Solovenia
Tahun lahir: 2002
Status (Aktif/Tidak Aktif/Free Agent): Aktif
Driver berhasil ditambahkan!
```

Gambar 4.4 Output Create Pembalap

```
=== LIST DRIVER ===
No  Nama                     Negara         Tahun Lahir Umur    Status
===========================================================================
1   Fernando Alonso          Spain          1981        45      Aktif
2   Lewis Hamilton           UK             1985        41      Aktif
3   Nico Hulkenberg          Germany        1987        39      Aktif
4   Valtteri Bottas          Finland        1989        37      Aktif
5   Sergio Perez             Mexico         1990        36      Aktif
6   Carlos Sainz             Spain          1994        32      Aktif
7   Alexander Albon          Thailand       1996        30      Aktif
8   Esteban Ocon             France         1996        30      Aktif
9   Pierre Gasly             France         1996        30      Aktif
10  Charles Leclerc          Monaco         1997        29      Aktif
11  Max Verstappen           Netherlands    1997        29      Aktif
12  George Russell           UK             1998        28      Aktif
13  Lance Stroll             Canada         1998        28      Aktif
14  Lando Norris             UK             1999        27      Aktif
15  Oscar Piastri            Australia      2001        25      Aktif
16  Liam Lawson              New Zealand    2002        24      Aktif
17  Franco Colapinto         Argentina      2003        23      Aktif
18  Isack Hadjar             France         2003        23      Aktif
19  Arvid Lindblad           Sweden         2004        22      Aktif
20  Gabriel Bortoletto       Brazil         2004        22      Aktif
21  Oliver Bearman           UK             2005        21      Aktif
22  Kimi Antonelli           Italy          2006        20      Aktif
```

Gambar 4.5 CRUD list pembalap sort berdasarkan umur(Descending)

```
=== LIST DRIVER ===
No  Nama                     Negara         Tahun Lahir Umur    Status
===========================================================================
1   Franco Colapinto         Argentina      2003        23      Aktif
2   Oscar Piastri            Australia      2001        25      Aktif
3   Gabriel Bortoletto       Brazil         2004        22      Aktif
4   Lance Stroll             Canada         1998        28      Aktif
5   Valtteri Bottas          Finland        1989        37      Aktif
6   Pierre Gasly             France         1996        30      Aktif
7   Esteban Ocon             France         1996        30      Aktif
8   Isack Hadjar             France         2003        23      Aktif
9   Nico Hulkenberg          Germany        1987        39      Aktif
10  Kimi Antonelli           Italy          2006        20      Aktif
11  Sergio Perez             Mexico         1990        36      Aktif
12  Charles Leclerc          Monaco         1997        29      Aktif
13  Max Verstappen           Netherlands    1997        29      Aktif
14  Liam Lawson              New Zealand    2002        24      Aktif
15  Fernando Alonso          Spain          1981        45      Aktif
16  Carlos Sainz             Spain          1994        32      Aktif
17  Arvid Lindblad           Sweden         2004        22      Aktif
18  Alexander Albon          Thailand       1996        30      Aktif
19  George Russell           UK             1998        28      Aktif
20  Lewis Hamilton           UK             1985        41      Aktif
21  Lando Norris             UK             1999        27      Aktif
22  Oliver Bearman           UK             2005        21      Aktif
```

Gambar 4.6 List sort berdasarkan Negara

```
=== LIST DRIVER ===
No  Nama                     Negara         Tahun Lahir Umur    Status
================================================================================
1   Kimi Antonelli           Italy          2006        20      Aktif
2   Oliver Bearman           UK             2005        21      Aktif
3   Arvid Lindblad           Sweden         2004        22      Aktif
4   Gabriel Bortoletto       Brazil         2004        22      Aktif
5   Isack Hadjar             France         2003        23      Aktif
6   Franco Colapinto         Argentina      2003        23      Aktif
7   Liam Lawson              New Zealand    2002        24      Aktif
8   Oscar Piastri            Australia      2001        25      Aktif
9   Lando Norris             UK             1999        27      Aktif
10  Lance Stroll             Canada         1998        28      Aktif
11  George Russell           UK             1998        28      Aktif
12  Max Verstappen           Netherlands    1997        29      Aktif
13  Charles Leclerc          Monaco         1997        29      Aktif
14  Alexander Albon          Thailand       1996        30      Aktif
15  Pierre Gasly             France         1996        30      Aktif
16  Esteban Ocon             France         1996        30      Aktif
17  Carlos Sainz             Spain          1994        32      Aktif
18  Sergio Perez             Mexico         1990        36      Aktif
19  Valtteri Bottas          Finland        1989        37      Aktif
20  Nico Hulkenberg          Germany        1987        39      Aktif
21  Lewis Hamilton           UK             1985        41      Aktif
22  Fernando Alonso          Spain          1981        45      Aktif
```

Gambar 4.7 Output sort berdasarkan tahun lahir

```
=== LIST DRIVER ===
No  Nama                    Negara         Tahun Lahir Umur    Status
===============================================================================
1   Alexander Albon         Thailand       1996        30      Aktif
2   Arvid Lindblad          Sweden         2004        22      Aktif
3   Carlos Sainz            Spain          1994        32      Aktif
4   Charles Leclerc         Monaco         1997        29      Aktif
5   Esteban Ocon            France         1996        30      Aktif
6   Fernando Alonso         Spain          1981        45      Aktif
7   Franco Colapinto        Argentina      2003        23      Aktif
8   Gabriel Bortoletto      Brazil         2004        22      Aktif
9   George Russell          UK             1998        28      Aktif
10  Isack Hadjar            France         2003        23      Aktif
11  Kimi Antonelli          Italy          2006        20      Aktif
12  Lance Stroll            Canada         1998        28      Aktif
13  Lando Norris            UK             1999        27      Aktif
14  Lewis Hamilton          UK             1985        41      Aktif
15  Liam Lawson             New Zealand    2002        24      Aktif
16  Max Verstappen          Netherlands    1997        29      Aktif
17  Nico Hulkenberg         Germany        1987        39      Aktif
18  Oliver Bearman          UK             2005        21      Aktif
19  Oscar Piastri           Australia      2001        25      Aktif
20  Pierre Gasly            France         1996        30      Aktif
21  Sergio Perez            Mexico         1990        36      Aktif
22  Valtteri Bottas         Finland        1989        37      Aktif
```

4.8 List Sort berdasarkan Urutan nama (A-Z)

```
=====================================================================================
DATA PEMBALAP LENGKAP
=====================================================================================
+----+--------------------------+----------------+-------------+-------+------------+
| No | Nama                     | Negara         | Tahun Lahir | Umur  | Status     |
+----+--------------------------+----------------+-------------+-------+------------+
| 1  | Charles Leclerc          | Monaco         | 1997        | 29    | Aktif      |
| 2  | Lewis Hamilton           | UK             | 1985        | 41    | Aktif      |
| 3  | George Russell           | UK             | 1998        | 28    | Aktif      |
| 4  | Kimi Antonelli           | Italy          | 2006        | 20    | Aktif      |
| 5  | Lando Norris             | UK             | 1999        | 27    | Aktif      |
| 6  | Oscar Piastri            | Australia      | 2001        | 25    | Aktif      |
| 7  | Fernando Alonso          | Spain          | 1981        | 45    | Aktif      |
| 8  | Lance Stroll             | Canada         | 1998        | 28    | Aktif      |
| 9  | Esteban Ocon             | France         | 1996        | 30    | Aktif      |
| 10 | Oliver Bearman           | UK             | 2005        | 21    | Aktif      |
| 11 | Isack Hadjar             | France         | 2003        | 23    | Aktif      |
| 12 | Max Verstappen           | Netherlands    | 1997        | 29    | Aktif      |
| 13 | Carlos Sainz Jr.         | Spain          | 1994        | 32    | Aktif      |
| 14 | Alexander Albon          | UK             | 1996        | 30    | Aktif      |
| 15 | Valtteri Bottas          | Finland        | 1989        | 37    | Aktif      |
| 16 | Sergio Perez             | Mexico         | 1990        | 36    | Aktif      |
| 17 | Nico Hulkenberg          | Germany        | 1987        | 39    | Aktif      |
| 18 | Gabriel Bortoleto        | Brazil         | 2004        | 22    | Aktif      |
| 19 | Pierre Gasly             | France         | 1996        | 30    | Aktif      |
| 20 | Franco Colapinto         | Argentina      | 1999        | 27    | Aktif      |
| 21 | Liam Lawson              | New Zealand    | 2002        | 24    | Aktif      |
| 22 | Arvid Lindblad           | Sweden         | 2004        | 22    | Aktif      |
| 23 | Benjamin Seskowi         | Solovenia      | 2002        | 24    | Tidak Aktif |
+----+--------------------------+----------------+-------------+-------+------------+
Pilih nomor pembalap yang ingin dihapus : 23
Pembalap berhasil dihapus.
```

4.9 Output Delete Pembalap

```
| 22 | Arvid Lindblad           | Sweden         | 2004        | 22    | Aktif      |
+----+--------------------------+----------------+-------------+-------+------------+
Pilih nomor pembalap yang ingin diedit : 22
Nama baru        : Arvid Limbad
Negara baru      : Swedia
Tahun lahir baru : 2007
Pilih status baru:
1. Aktif
2. Tidak Aktif
Pilihan          : 1
Pembalap berhasil diedit.
```

Gambar 4.10 Edit Pembalap

```
1. List Driver
2. Tambah Driver
3. Edit Driver
4. Hapus Driver
5. List Kontrak
6. Tambah / Transfer Kontrak
7. Logout
Pilih: 5

=== SORT KONTRAK BY ===
1. Nama Driver (A-Z)
2. Nama Tim (A-Z)
3. Tahun Mulai (Descending)
4. Tahun Selesai (Descending)
Pilih: 1

=== TAMPILKAN KONTRAK ===
1. Semua kontrak
2. Hanya kontrak aktif tahun 2026
Pilih: 1

=== LIST SEMUA KONTRAK ===
No  Driver                   Tim                 Mulai       Selesai     Status 2026
=========================================================================================
1   Alexander Albon          Ferrari             2025        2027        Aktif
2   Arvid Lindblad           Ferrari             2025        2027        Aktif
3   Carlos Sainz             Mercedes            2025        2027        Aktif
4   Charles Leclerc          Mercedes            2025        2027        Aktif
5   Esteban Ocon             McLaren             2025        2027        Aktif
6   Fernando Alonso          McLaren             2025        2027        Aktif
7   Franco Colapinto         Aston Martin        2025        2027        Aktif
8   Gabriel Bortoletto       Aston Martin        2025        2027        Aktif
```

Gambar 4.11 Output List Kontrak

5. Langkah-langkah GIT

```
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git add .
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git commit -m "KOMAT KAMIT"
[main aede32e] KOMAT KAMIT
 3 files changed, 233 insertions(+)
 create mode 100644 post-test/post-test-7/2509106103-KhayzanDwitraAttala-PT-7 (1).pdf
 create mode 100644 post-test/post-test-7/2509106103-PT-7.drawio.png
 create mode 100644 post-test/post-test-7/2509106103_KhayzanDwitraAttala-PT-7.py
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7> git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 868.02 KiB | 27.13 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Khayzannn/praktikum-apd.git
   609b85f..aede32e  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Asus GK\OneDrive\Documents\praktikum-apd\post-test\post-test-7>
```

Gambar 5.1 Langkah - Langkah Git

5.1 GIT Add

Git add adalah perintah diGit yang digunakan untuk menambahkan perubahan pada berkas di direktori kerja ke area staging (atau indeks), mempersiapkannya untuk dimasukkan ke dalam commit berikutnya.

5.2 GIT Commit

Git commit adalah perintah di sistem kontrol versi Git untuk menyimpan (melakukan "commit") serangkaian perubahan yang telah dipilih (di-staging) ke dalam repositori lokal Anda, membuat sebuah "snapshot" atau titik pemeriksaan pada riwayat proyek dengan deskripsi perubahan.

5.3 GIT Push

Git push adalah perintah dalam Git yang digunakan untuk mengunggah (mengirimkan) perubahan yang telah dicatat di repositori lokal Anda ke repositori jarak jauh (seperti GitHub), sehingga memperbarui cabang tersebut dan membagikan pekerjaan dengan tim atau menyinkronkan dengan versi pusat.